BCA 2nd Sem.

CC-104 (R)

2025

# Data Structure

Full Marks: 100 Time: 3.00 Hours

1. **Answer *all* questions.** [1 X 10]

a. Give an example of nonlinear data structure.
b. Define ADT.
c. Give an example of divide-and-conquer algorithm.
d. Give a real time example of single linked list in your daily life.
e. What is the use of hash function in data structure?
f. What is the prerequisite conditions for binary search?
g. A binary tree contains at most ------ number of child nodes.
h. How many nodes are required to represent a polynomial equation using linked list?
i. When a graph is said to be a tree?
j. Which data structure is more appropriate for implementing quick sort iteratively?

2. **Answer *all* questions.** [2 X 9]

a. Write two applications of stack.
b. Convert (A+B/C*D)-F+E into postfix form.
c. What is AVL tree?
d. Give a pictorial representation of doubly linked list.
e. Mention the operations that can be carried out in data structure.
f. What is the difference between enqueue () and dequeue ()?
g. What are the different types of tree traversing?
h. Define Strictly binary tree?
i. What do you mean by internal and external sorting?

3. **Answer any *Eight*.** [5 X 8]

a. Write an algorithm to delete a node from a doubly linked list.
b. Write an algorithm to implement linear search.
c. What is circular linked list? Explain.
d. Evaluate the following postfix expression showing the stack status.
P: 3, 5, +, 6, 4, -, *, 4, 1, -, 2, ^, +
e. What is hashing? Explain the problems of using hashing function.
f. Differentiate between merge sort and quick sort.
g. What is recursion? Write a C program to find the GCD of two positive numbers using recursion.

h. What is the difference between calloc() and malloc()?
i. Explain bubble sort.
j. How do you resolve the collision problem using chaining method? Explain.

**4. Answer any Four.** **[8 X 4]**

a. Construct an expression tree for the expression (a + b * c) + ((d * e + f) * g). Write the outputs by applying preorder, inorder and postorder traversals.
b. What is data structure? Briefly explain various data structures with valid examples.
c. How do you represent two dimensional array in the memory? Explain.
d. Define a linked list. Explain different types of it and outline the advantages and disadvantages of each type.
e. Define binary search. Search the element 60 from the following list using binary search.
L: 5, 10, 15, 18, 35, 40, 48, 60, 69, 80.
0 1 2 3 4 5 6 7 8 9

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b * c ) + { ( d * e + f ) * g }
( post order )

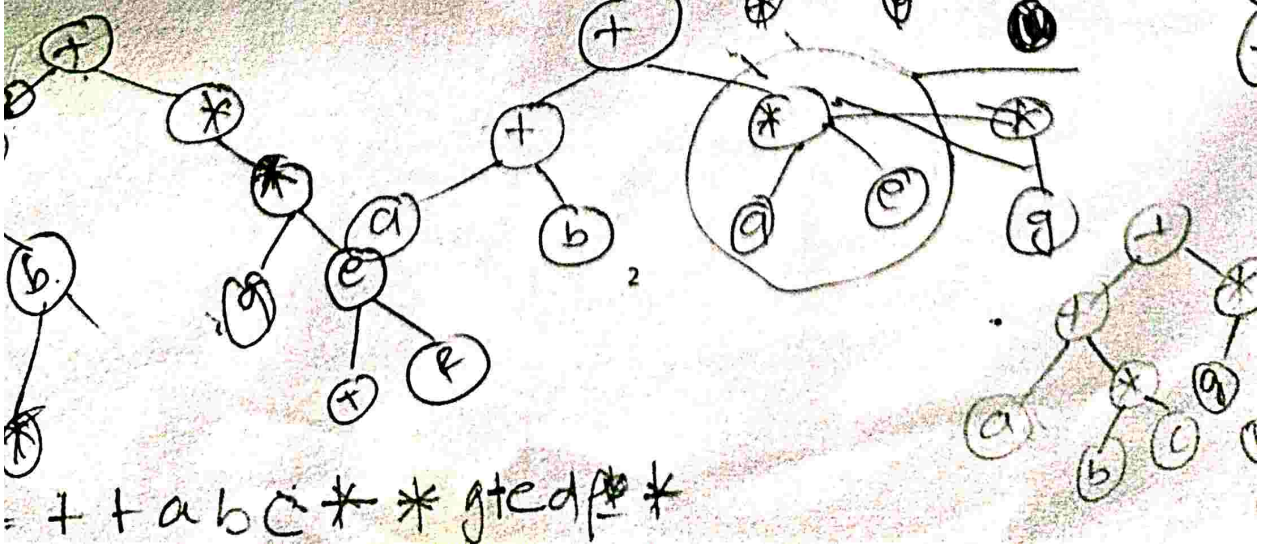
Pre order

ab + c * + ( de * + fg * .

Pre a + b * c + g * e

+ c * de * fg * + +

+ * + ab c * de * fg



+ + abc * * gted * *

BCA 2nd Sem.
CC-103 (R)

2025

# Mathematics Foundation to Computer Science-II

Full Marks- 100 Time : 3.00 Hours

The figures in the right hand margin indicate Marks. *Symbols carry usual meaning.*

**Q1 Answer all questions.** [1 × 10]

(a) Is the proposition (p v q)→q form a tautology?
(b) Suppose n is nonnegative integer. Then determine the value of $1 + 2 + 3 + \cdots + n$.
(c) Let $p(x)$ denote the statement "$x > 3$". Then determine the truth value of $p(4)$ and $p(2)$.
(d) In a cyclic group, every element is a generator of it. (True/ False).
(e) Give an example of a nonempty set, which is a semi-group but not a group.
(f) Show that a group with no proper non-trivial subgroup is cyclic.
(g) Write 2 limitations of Newton's method.
(h) Find the unique polynomial of degree $\leq 2$, such that $(0) = 1, f(1) = 3, f(3) = 5$.
(i) While solving a transportation problem, what is the meaning of occurrence of degeneracy?
(j) While solving a linear programming problem by simplex method, how can you know about the existence of unbounded solutions?

**Q2 Answer all questions.** [2 × 9]

(a) Explain the terms "antecedent" and "Consequent" with suitable counter examples.
(b) What rule of inference is used in this argument? If I study hard, then I get A's. I study hard. Therefore I get A's.
(c) In a group $G$ prove that $(ab)^{-1} = b^{-1}a^{-1}$ for all $a, b \in G$.
(d) Prove that in a group $G$, there is only one identity element.
(e) Show that $Z_8$ is cyclic.
(f) Write the limitations of Lagrange's interpolation method.
(g) When does the trapezoidal rule for integrating $\int_a^b f(x)dx$ gives exact results?
(h) Write two limitations of linear programming problems (LPP).
(i) What do you mean by infeasible solution?

**Q3 Answer any Eight.** [5 × 8]

(a) Define logical equivalence. Show that $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$ are logically equivalent.
(b) Use mathematical induction to prove that $n^3 - n$ is divisible by 3 whenever $n$ is a positive integer.

(c) Explain the convergence of Newton's method.

(d) Using Lagrange interpolation, find y(10) given that y(5) = 12, y(6) = 13, y(9) = 14, y(11) = 16.

(e) Find the approximate value of $I = \int_0^{1.2} \frac{dx}{1+x}$ by Simpson's $\frac{1}{3}rd$ rule.

(f) Let $G = \left\{\begin{bmatrix} a & a \\ a & a \end{bmatrix} : a \in R\right\}$. Show that $G$ is a group under multiplication and find the identity element.

(g) Prove that a group of order 4 is always abelian.

(h) Show that the set of all roots of $x^6 = 1$ form a group with respect to multiplication.

(i) Use Vogel's approximation method find the initial basic feasible solution for the following transportation problem:

$$\begin{array}{c} \\ A \\ B \\ C \\ \\ \end{array} \begin{array}{c} \begin{array}{cccc} D & E & F & G \end{array} \\ \begin{bmatrix} 11 & 13 & 17 & 14 \\ 16 & 18 & 14 & 10 \\ 21 & 24 & 13 & 10 \end{bmatrix} \\ \begin{array}{cccc} 200 & 225 & 275 & 250 \end{array} \end{array} \begin{array}{l} \\ 250 \\ 300 \text{ Available} \\ 400 \\ \\ \end{array}$$

Demand

(j) Obtain the initial basic feasible solution for the following transportation problem using north-west corner rule:

$$\begin{array}{c} \\ A \\ B \\ C \\ \\ \end{array} \begin{array}{c} \begin{array}{cccc} D & E & F & G \end{array} \\ \begin{bmatrix} 5 & 3 & 6 & 2 \\ 4 & 7 & 9 & 1 \\ 3 & 4 & 7 & 5 \end{bmatrix} \\ \begin{array}{cccc} 16 & 18 & 31 & 25 \end{array} \end{array} \begin{array}{l} \\ 19 \\ 37 \text{ Available} \\ 34 \\ \\ \end{array}$$

Demand

**Q 4 Answer any Four.** **[8 × 4]**

(a) Write short notes on Conjunctive Normal forms and Disjunctive Normal forms.

(b) Evaluate $\int_0^1 \frac{dx}{1+x^2}$ using Trapezoidal rule with h=0.2. Hence obtain an approximate value of $\pi$.

(c) Obtain the initial basic feasible solution for the following transportation problem. Is it optimal one?

$$\begin{array}{c} \\ A \\ B \\ C \\ \\ \end{array} \begin{array}{c} \begin{array}{cccc} D & E & F & G \end{array} \\ \begin{bmatrix} 5 & 4 & 2 & 1 \\ 2 & 3 & 7 & 5 \\ 5 & 4 & 5 & 6 \end{bmatrix} \\ \begin{array}{cccc} 40 & 50 & 70 & 100 \end{array} \end{array} \begin{array}{l} \\ 130 \\ 100 \text{ Available} \\ 30 \\ \\ \end{array}$$

Demand

(d) State and prove Lagrange's theorem.

(e) When an alternative optimal solution will exist for an LPP. Solve the LPP Maximize $z = 4x_1 + 10x_2$ subject to the constraints: $2x_1 + x_2 \leq 50, 2x_1 + 5x_2 \leq 00, 2x_1 + 3x_2 \leq 90; x_1, x_2 \geq 0$ by simplex method. Determine the alternative optimal solution if any.

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BCA 2nd Sem.

CC – 105 (R)

June-2025

# Operating Systems

Full Marks – 100

Time : 3.00 Hours

1) **Answer all questions.** [1 X 10 ]

i) Define Operating System.
ii) What is the function of the Process Control Block?
iii) Name one Pre-emptive Scheduling Algorithm.
iv) What is a semaphore used for?
v) What is the purpose of the Banker's Algorithm?
vi) Define fragmentation.
vii) What is demand paging
viii) Mention any one advantage of multithreading.
ix) What is a critical section?
x) Name a disk scheduling algorithm.

2) **Answer all questions.** [ 2 X 9]

i) Differentiate between multiprogramming and multitasking.
ii) Define threads and explain the benefits of using them.
iii) What are performance measures for Process Scheduling algorithms.
iv) What are race conditions in process synchronization?
v) Explain any one method of deadlock prevention.
vi) What is internal fragmentation?
vii) List the components of an Operating System.
viii) Describe the concept of time-sharing systems.
ix) What is Thrashing?

) **Answer any Eight.** [ 5 x 8]

i) Explain the different types of Operating Systems with examples.
ii) Describe the architecture and components of an Operating System.
iii) Explain the process states and their transitions with a diagram.
iv) Discuss multithreading and its benefits.
v) Explain Round Robin Scheduling with a suitable example.
vi) Define semaphores and explain how they help in mutual exclusion.
vii) Describe the deadlock detection and recovery techniques.
viii) What is paging? Explain with an example.
ix) Explain FIFO and LRU Page Replacement Algorithms.
x) Discuss LOOK and SCAN disk scheduling algorithms with examples.

Q4) **Answer any Four.** [4 x 8]

i) **What is Process Synchronization?** Explain the Critical Section Problem and describe solutions using Semaphores. How do synchronization tools help avoid these issues?

ii) What are various CPU schedulers? Differentiate between preemptive and non-preemptive Process scheduling algorithms, Give one example from each.

iii) What is Demand paging? Describe the process of how can demand paging be implemented with virtual memory?

iv) What are the necessary conditions for Deadlock?

Consider the following snapshot of a system

Allocation

| Process | A | B | C |
|---|---|---|---|
| $P_0$ | 1 | 1 | 1 |
| $P_1$ | 2 | 0 | 0 |
| $P_2$ | 3 | 0 | 2 |
| $P_3$ | 2 | 1 | 1 |
| $P_4$ | 0 | 0 | 2 |

Max

| A | B | C |
|---|---|---|
| 7 | 5 | 3 |
| 3 | 2 | 2 |
| 4 | 0 | 2 |
| 2 | 2 | 2 |
| 4 | 3 | 3 |

Available

| A | B | C |
|---|---|---|
| 3 | 3 | 2 |

Using banker's algorithm, find out if the above system is in safe state. If yes, find the safe sequence.

v) **Write short notes on the following (any two):**

a) System Calls

b) Operating System Structure

d) Real-Time Operating Systems

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2025

# Indian Constitution

Full Marks: 100 Time: 3.00 Hours

**1. Answer all questions.** [1X10]

(A) What is the short title of the Indian Constitution?
(i) The Indian Constitution (ii) The Constitution of India
(iii). The Constitutional Law (iv) None of the above

(B) How many fundamental duties are there in the Indian Constitution?
(i) 10 (ii) 11 (iii) 15 (iv) 17

(C) The executive power of the Union is vested in-
(i) President of India (ii) Chief Justice of India
(iii). Chief Election Commissioner (iv). None of the above

(D) The first Prime Minister of India was-
(i) Lal Bahadur Shastri (ii) Morarji Desai
(iii) Pandit Jawaharlal Nehru (iv) None of the above

(E) The Governor of a State is appointed by-
(i) Chief Justice of India (ii) President of India
(iii) Prime Minister (iv)None of the above

(F) Ministers hold office during the pleasure of-
(i) Chief Justice (ii) Chief Minister
(iii) Governor (iv)None of the above

(G) Constitution (Seventy-third Amendment) Act was enacted in the year-
(i) 1985 (ii)1990 (iii) 1992 (iv) 1998

(H) Which of the following constitution amendments is related to urban local government?
(i) 73rd Constitution amendment (ii) 74th Constitution amendment
(iii) 75th Constitution amendment (iv) 76th Constitution amendment

(I) The Election Commission of India is a constitutional body. This statement is-
9i) Correct (ii) Partly correct (iii) Incorrect (iv)Partly incorrect

(J) The State Election Commission, Odisha came into existence in the year-
(i) 1990 (i) 1992 (iii) 1994 (iv) 1996

[2x9]

**2. Answer all questions.**

(i) What is the basic structure of the Constitution?
(ii) What do you mean by fundamental rights?
(iii) Write the composition of Lok Sabha.
(iv) What is ordinance making power of the President?
(v) Write about the state secretariat.
(vi) What are the qualifications for appointment as Governor?
(vii) What do you understand by local administration?
(viii) Write the composition of Zila Panchayat.
(ix) Write any two functions of the State Election Commission.

[5X8]

**3. Answer any eight questions.**

(i) Write a note on the history of making of Indian Constitution.
(ii) What are the Directive Principles of State Policy?
(iii) Write the structure of the Indian Union.
(iv) What are the functions of Rajya Sabha?
(v) Chief Minister is the real head of Executive. Explain.
(vi) What are the functions of state secretariat?
(vii) Write the significance of the Municipal Corporation.
(viii). What are the main functions of the district administration?
(ix). Who can be appointed as the Chief Election Commissioner?
(x). Mention the key functions of the Election Commission of India.

**4. Answer any four questions.** [8X4]

(i) What are the fundamental duties as given in the Indian Constituti
Mention its significance also.
(ii) Discuss different powers of the President of India.
(iii) Explain the role and powers of the Governor.
(iv) Discuss the functions of the Municipal Corporation with examples.
(v) Discuss the role of the Election Commission of India.